

LAB 03

→ Objectives

The objective of this lab is to introduce you to dynamic memory allocation and strings in C programming.

- 1) Go over the solution for Lab02
- 2) Illustrate how to use dynamic memory allocation and strings in C

→ Exercise

Task 1

Create a function to perform the insertion sort. Now the program will perform the following steps:

- ☒ Prompt the user to enter the number of array elements (say, N)
- ☒ Read the number of elements (N)
- ☒ Use dynamic memory allocation to allocate an array:
 - a) N single precision floating-point elements (C type float).
- ☒ Read the N single precision floating-points elements to the allocated array
- ☒ Invoke a function to sort the array using insertion sort ...
 - a) insertion sort function should take the array reference and number of elements as arguments and sort the array in-place
 - b) insertion sort function should have the return type as void, i.e. it does not return anything
- ☒ Print the sorted array

Task 2

In C, there is no primitive datatype called string. Strings are implemented as an array of characters typically terminated with the null character (`\0`). For example:

```
char hello[6] = {'H', 'e', 'l', 'l', 'o', '\0'};
```

// OR

```
char hello[] = "Hello\0";
```

NOTE: When you initialize an array of characters, it is a good idea to add the null character – `\0` as the terminating character.

Now let's see a few helpful functions for character array manipulation. For this action you need to include the header file called "string.h".

strlen: to get the length of string

strcpy: to copy one string to another

strcat: to concatenate one string to another

strcmp: to compare one string to another string. This will return integer value based on the two strings compared. The return value is 0 if the two strings are equal, greater than 0 if first string is greater than the second string, and less than 0 if the first string is less than the second string. The comparison is performed using unsigned characters.

strchr: to search first occurrence of a character in the given string. This will return either a pointer to the first occurrence or NULL if not found.

strstr: to search first occurrence of a sub-string in a given string. This will return either a pointer to the first occurrence of the sub-string or NULL if not found.

Looking at some of these function? You can find out more about these function by typing:

☒ use the 'man <functionname>' to find out more about one of the str functions

☒ implement following example for string manipulations:

- strcat(hello, name);
- printf("%s\n", hello);

NOTE: the strcat function concatenates second string (name) into first string (hello), also notice that hello has one extra space before string termination to print "hello <name>" with space between "hello" and <name>.

☒ utilize the 'strcmp' function to gain a better understanding of how the function works

- the 'strcmp' function performs a character-by-character lexicographic comparison to compare the two strings.

→ Assignment

☒ modify the insertion sort program developed in Lab02:

- I) program now sorts strings (array of characters)
- II) uses dynamic memory allocation to allocate memory based on the number of elements in the array as well as the length of individual strings

NOTE: *scanf* to read the strings, determine the length of the string, and then allocate appropriate amount of memory.

☒ use separate functions to →

- a) read the strings
- b) sort the strings
- c) display the sorted strings

☒ use appropriate string functions provided by the C library